

Vinay Prakash

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EXPERIENCE

Robotics HW-SW Integration Engineer (R&D Team), Halo.car

Oct 2024 - Present

- Optimized HW-SW integration for automotive retrofit kits, reducing system commissioning time by 67% (6 to 2 hrs).
- Developed a tool with multi-format log parsing, synced video, dynamic bitfield display & signal plotting to reverse-engineer Radar CAN frames; fused camera data for L2 autonomy (LKA, FCW, ACC, AEB).
- Modified and enhanced Linux drivers for force feedback systems, emulating realistic on-road haptic driving experiences.
- Reverse-engineered vehicle controls enabling driverless remote piloting projects with Nellis AFB, Uber, and Audi.
- Implemented calibration routines for vehicle steering, brakes, and throttle control via a custom-built utility tool.
- Architected a HW/SW diagnostics and error-handling pipeline for a limp-home mode to mitigate critical system failures.

Robotics Integration Engineer (Autonomy Team), Monarch Tractor

Nov 2022 - Oct 2024

- Maintained and refactored the SW-FW interface layer for 4 CAN buses on NVIDIA Orin CAN interfaces for autonomy.
- Devised CAN interface bringup, diagnostics and error handling SW policies from prototype to production tractors.
- Delivered a metrics and diagnostic pipeline for value-added insights for customer about operations and tractors.
- Executed the HRI lights project for FW and SW development compliant with OSHA regulations for Ag environment.
- Led several internal projects like Front Loader implements and ISOBUS abstraction layer for autonomous operations.

Electrical Engineer (Electrical Engineering Team), Monarch Tractor

Jun 2021 - Oct 2022

- Managed electrical component sizing, sourcing, design, testing & validation for 5 out of 8 electrical sub-systems.
- Migrated 2D electrical harness schematics from Visio to SW Electrical - created 2D electrical schematics, wiring and component libraries, 3D electrical component libraries, worked with mechanical engineers to route harnesses in 3D.
- Designed custom report formats in SW Electrical for BOM, To-From lists, label and wire/cable cut lists, formboard.
- Trained the team on the SW Electrical and resolved wire harness related design issues for all 8 electrical sub-systems.
- Responsible for HW-SW integration and testing for electrical components from prototype to production models.
- Performed root cause analysis and troubleshooting for electrical failures in a fleet of 300+ tractors, ensuring minimal downtime and increased system reliability.

Electronics R&D Engineer (Hardware Engineering Team), Lithion Power Pvt. Ltd

Dec 2017 - May 2018

- Developed data logging solution for monitoring real-time battery systems via Thingspeak and AWS-based dashboards.
- Migrated data logger to ARMv8-M in 8 months for production with PCB bring-up, debugging, and validation/diagnostics.
- Integrated data logger in BMS for remote diagnostics with superuser control features: geo-fencing, and remote shutdown.
- Implemented SoC and SoH calculation-based drive feedback for drivers and superusers via texts and dashboards.

PROJECTS

Research Assistant (Connected & Autonomous Vehicles), MTU

- Project 1 - Novel applications of machine learning algorithms for IEEE 802.11 for wireless sensing.
- Project 2 - Performance evaluation of 5G based communication for V2X non-safety applications for CAVs.
- Project 3 - ECU-to-ECU communication performance evaluation using 5G for V2X safety applications.

Autonomous-AI-Racecar

- Migrating the NVIDIA JetRacer project from Jetson Nano to Jetson Orin, integrating OAK D-Lite & RPi cameras for driveable area segmentation, waypoint generation and tracking.
- Implemented CAN bus motor control using a PIC18F MCU with a custom SocketCAN-ROS interface.
- Developing a reinforcement learning-based driving controller using Isaac Sim and Isaac Lab simulation environments.

EDUCATION

Master of Science, Electrical and Computer Engineering

Michigan Technological University

Aug 2019 - Apr 2021

GPA: 3.5

Bachelor of Technology, Electronics and Communication Engineering

Rajasthan Technical University

Aug 2013 - Jun 2017

Grade: First Division

SKILLS

SW & Engineering Applications: MATLAB, SolidWorks Electrical, E3 series, RapidHarness, IAR Workbench, Keil uVision, Bash, Ghidra, Jenkins, Python, C, C++, ROS, ROS 2

uC's & FW Development: ATmega, ARM, PIC, STM32, ESP32, Nuvoton, SIMCom, Telit, uBlox, RTOS, Bootloaders

HW Design & Validation: Altium, EAGLE, NI MultiSim, Oscilloscope, Logic Analyzer, Multimeter, Signal Generator

Networking & Communication Protocols: CAN, LIN, ISOBUS, OBD-II, Ethernet, SPI, I2C, UART, TCP/IP, UDP, MQTT, DDS, GSM, Wireshark, Vector CAN tools, PCAN View, PiSnoop

Simulation & Dev Tools: Docker, Git, RViz, Gazebo, Foxglove, Isaac Sim, Isaac Lab, Carla, NS-3, SUMO, VISSIM

Project Management & Collaboration: Agile Methodologies, JIRA, Confluence, Jama, Asana, GitHub, Linear